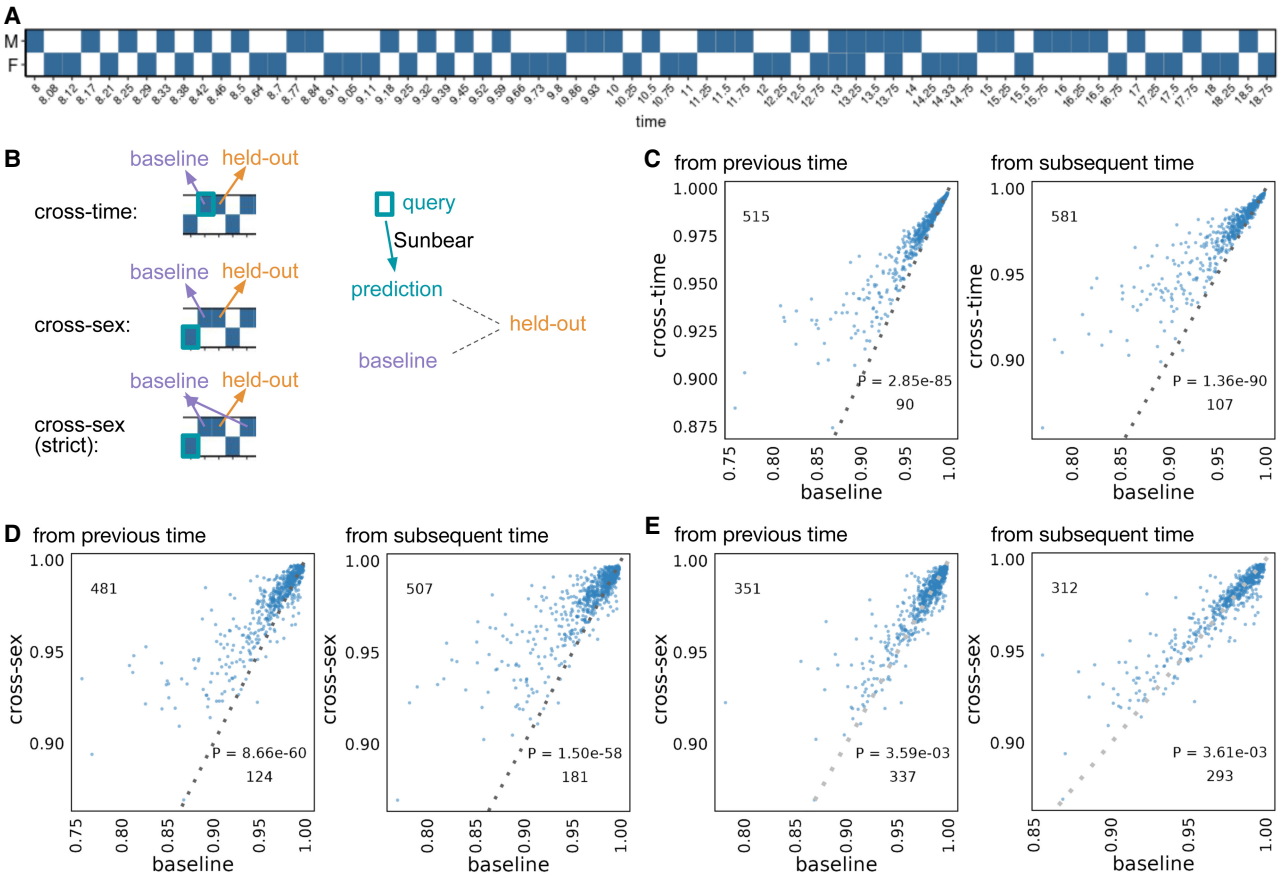

Figure 2. Single-cell profile inference across time and conditions. (*A*) Sunbear is trained on scRNA-seq profiles of whole-mouse embryos collected at alternating sexes along developmental time points (Qiu et al. 2024). (*B*) Sunbear is validated in three scenarios. In each scenario, one data block is held out from the training, and Sunbear is used to predict the profile of the missing block based on cells in the query block (outlined in turquoise). Sunbear's prediction is compared against the baselines using the held-out block's nearest existing measurements with the desired sex factors. The pseudobulk Pearson's correlation per major cell trajectory is calculated between the held-out profile and predictions/baselines. (*C*) Cross-time evaluation: query and baseline are selected either from the closest previous time point (*left*) or from the closest subsequent time point (*right*). The pseudobulk Pearson's correlation between the original held-out profile and predicted (*y*-axis) and baselines (*x*-axis) is plotted for each major cell trajectory in each held-out time point. Each dot represents a cell trajectory per held-out time point, and numbers indicate the number of dots *above* and *below* the diagonal line. *P*-values are calculated by a one-sided Wilcoxon rank-sum test. (*D*) Cross-sex prediction: similar to that in *C*, except query cells are selected from the opposite sex to the held-out data. (*E*) Cross-sex prediction: similar to that in *D*, except we enforce a strict baseline model by taking the mean of the previous and subsequent time point per cell trajectory.

methods in the majority of cell types and time points (Fig. 2B, "cross-time"; Fig. 2C).

In addition to modeling a continuous variable such as time, Sunbear is capable of modeling discrete factors corresponding to various types of experimental conditions. This functionality allows us to swap the condition factor to predict a cell's profile in a missing time point and in a different condition, thus making direct comparisons of gene expression profiles across sexes. We used this functionality to predict sex-specific single-cell profiles at missing time points (Fig. 2B, "cross-sex"). To evaluate our model's performance, we held out each time point and sex at a time, and we used profiles from the opposite sex to recapitulate the held-out profile. Sunbear is able to predict scRNA-seq profiles in an unseen time point in the opposite sex, with significant improvement compared with using baseline models of the previous and/or subsequent time points in the held-out sex (Fig. 2D,E).

To test whether Sunbear can operate on data sets with sparser time points, we performed two additional experiments: (1) iteratively holding out the target time point's neighboring one, two, or three time points (Supplemental Fig. S3) and (2) making the data sets sparser with gaps of 0.5, 0.75, and 1 day (Supplemental Fig. S4). Our result shows that the method is robust to data with sparser time points.

Furthermore, we compared Sunbear with TrajectoryNet on the sparser data set with a 0.75 day time gap. This analysis showed that Sunbear also outperforms TrajectoryNet in inferring the held-out time point profile (Supplemental Fig. S5; Supplemental Methods, "Comparison with TrajectoryNet").

We further used Sunbear to integrate three sets of single-cell RNA-seq time-series profiles (Cao et al. 2019; Pijuan-Sala et al. 2019; Qiu et al. 2022). To assess how well the integration worked, we quantified LISI scores (Korsunsky et al. 2019) between each pair of neighboring time points, as well as between data sets collected at matched time points. Our results suggest that Sunbear can accurately align single-cell time-series profiles across time points and batches (Supplemental Fig. S6).

Finally, we performed several ablation studies on different parts of the model. To assess the contribution of the sinusoidal encoding of time information, we compared Sunbear with a variant of Sunbear that encoded time factor as a single value representing